Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CLEAN AND GREEN TECHNOLOGY

The domain of the Project:

FULL STACK DEVELOPMENT

Project Done By:

N Sai Kiran - G30 FSD

Period of the project

March 2026 to April 2026

(12/03/2026 - 30/04/2026)

Declaration

The project titled "Clean and Green Technology" has been mentored by Ms.Akshara(Software Developer at Propel Technology), organised by SURE Trust, from March 2026 to April 2026, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge that I have worked on it successfully and enhanced the practical knowledge in the domain.

Member:

Mr. N Sai Kiran

Mentor's Name
Ms.Akshara
(Software Developer at Propel Technology)

Prof. Radhakumari
Executive Director & Founder
SURE Trust

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Executive Summary

This project focuses on the design and development of a **web-based system for efficient issue reporting and management**, aimed at improving communication between citizens and authorities. The platform enables users to report civic or disaster-related issues, track their status in real time, and ensure timely resolution through an organized administrative workflow.

The system is developed using modern web technologies, including Angular for the user interface, Django for backend logic and API development, and PostgreSQL for reliable data storage. The application is deployed using cloud platforms such as Vercel (frontend) and Render (backend), ensuring accessibility, scalability, and performance.

🔹 Objectives:

The primary objective is to build a **transparent, user-friendly, and scalable system** that allows:

- Easy submission of issues by users
- Real-time tracking of complaints
- Efficient management and resolution by administrators

🔹 Methodology:

The project follows a structured development approach:

- Requirement analysis and system design
- Frontend and backend development
- Database integration
- Testing and debugging
- Deployment on cloud platforms

🔹 Key Findings:

- The system significantly improves **response time and efficiency**
- Enables **real-time monitoring and centralized data management**
- Provides a **user-friendly interface** for both citizens and administrators
- Demonstrates the effectiveness of full-stack development in solving real-world problems

🔹 Recommendations:

- Integrate **location-based services (GPS)** for better issue tracking
- Implement **AI-based prioritization** of issues
- Develop a **mobile application** for wider accessibility
- Enhance security features and scalability for large-scale deployment

Background and Context of the Project

With the rapid growth of urban populations and increasing dependence on digital solutions, the need for efficient communication between citizens and governing authorities has become more critical than ever. Traditional systems for reporting civic issues or disaster-related incidents often rely on manual processes such as phone calls or in-person complaints, which are time-consuming and lack transparency.

To address these challenges, modern web technologies provide an opportunity to build centralized platforms that allow real-time reporting, monitoring, and resolution of issues. This project leverages technologies like Angular, Django, and PostgreSQL to create an efficient and scalable system.

The project is designed in the context of smart city initiatives and digital governance, aiming to enhance public service delivery and citizen engagement.

Problem Statement / Goals of the Project

Problem Statement:

Existing systems for reporting civic issues or managing disaster-related information suffer from several drawbacks:

- Lack of real-time tracking
- Inefficient communication channels
- Delayed response from authorities
- Poor data management and analysis
- Limited accessibility for users

Goals of the Project:

The primary goals of this project are:

- To develop a **user-friendly web platform** for reporting issues
- To enable **real-time tracking and updates**
- To improve **coordination between citizens and authorities**
- To ensure **transparency and accountability** in issue resolution
- To build a **scalable and reliable system** using modern technologies

🔹 Scope and Limitations of the Project

Scope:

The scope of the project includes:

- User registration and authentication
- Submission of issues with descriptions
- Admin dashboard for monitoring and managing complaints
- Status tracking (Pending, In Progress, Resolved)
- Cloud deployment for remote accessibility

The system can be extended to support features such as location tracking, image uploads, and notification systems.

Limitations:

- Limited to web-based access (no mobile app currently)
- Depends on internet connectivity
- Basic security implementation (can be enhanced further)
- Limited scalability in free hosting environments
- Does not include advanced AI-based analysis (future enhancement)

🔹 Innovation Component in the Project

This project introduces several innovative aspects compared to traditional systems:

- **Real-time Issue Tracking:** Users can monitor the status of their complaints instantly
- **Digital Transformation:** Replaces manual reporting with an online platform
- **User-Centric Design:** Simple and intuitive interface for easy accessibility
- **Scalable Architecture:** Built using modern frameworks that support future expansion
- **Cloud Deployment:** Ensures global accessibility and reliability using platforms like Vercel and Render

Additionally, the system can be further enhanced with technologies such as AI, geolocation services, and real-time notifications, making it adaptable for smart city and disaster management applications.

Clearly Defined Objectives and Goals of the Project

The primary objective of this project is to design and develop a **web-based application** that facilitates efficient reporting, tracking, and resolution of civic or disaster-related issues. The system aims to bridge the communication gap between citizens and authorities by providing a centralized and transparent platform.

The key objectives of the project are:

- To develop a **user-friendly interface** using Angular that allows users to easily report issues
- To implement a **robust backend system** using Django for handling business logic and API communication
- To ensure **secure and efficient data management** using PostgreSQL
- To enable **real-time tracking of issues**, allowing users to monitor the progress of their complaints
- To provide an **admin dashboard** for authorities to manage, update, and resolve issues effectively
- To ensure **transparency and accountability** in the complaint resolution process
- To build a **scalable and maintainable system** that can handle future enhancements

Expected Outcomes and Deliverables

Upon successful completion, the project is expected to deliver the following outcomes:

Functional Outcomes:

- A fully operational **web application** accessible through browsers
- A system that allows users to:
  - Register and log in
  - Submit issues with details
  - Track issue status (Pending, In Progress, Resolved)
- An admin interface to:
  - View reported issues
  - Update status and manage complaints

Technical Deliverables:

- Frontend application developed using Angular
- Backend REST API developed using Django
- Database schema implemented in PostgreSQL
- Integrated system with proper API communication
- Deployment on cloud platforms such as Vercel and Render

Documentation Deliverables:

- Complete project report (including diagrams and explanations)
- Source code with proper structure and comments
- GitHub repository link for version control
- Deployment URLs for live demonstration

Performance Outcomes:

- Faster issue reporting and resolution
- Improved communication between users and authorities
- Efficient data handling and retrieval
- Reliable system performance under normal usage conditions

🔹 Methods / Technology Used

The project follows a **full-stack web development methodology**, combining frontend, backend, and database technologies to build a scalable and efficient system.

- **Frontend Development:**
  The user interface is developed using Angular, which provides a component-based architecture for building responsive and dynamic web pages.
- **Backend Development:**
  The server-side logic is implemented using Django, which handles API creation, request processing, authentication, and business logic.
- **Database Management:**
  Data is stored and managed using PostgreSQL, ensuring reliability, consistency, and efficient querying.
- **Deployment Strategy:**
  The application is deployed using Vercel for the frontend and Render for the backend, enabling accessibility over the internet.

🛠️ Tools / Software Used

The following tools were used during development:

- **Code Editor:** Visual Studio Code
- **Version Control:** Git & GitHub
- **API Testing:** Postman
- **Browser Tools:** Chrome Developer Tools
- **Package Managers:** npm (for Angular), pip (for Django)
- **Database Tools:** pgAdmin / Supabase dashboard

These tools helped streamline development, testing, debugging, and deployment processes.

📊 Data Collection Approach

The system primarily collects **user-generated data**, which includes:

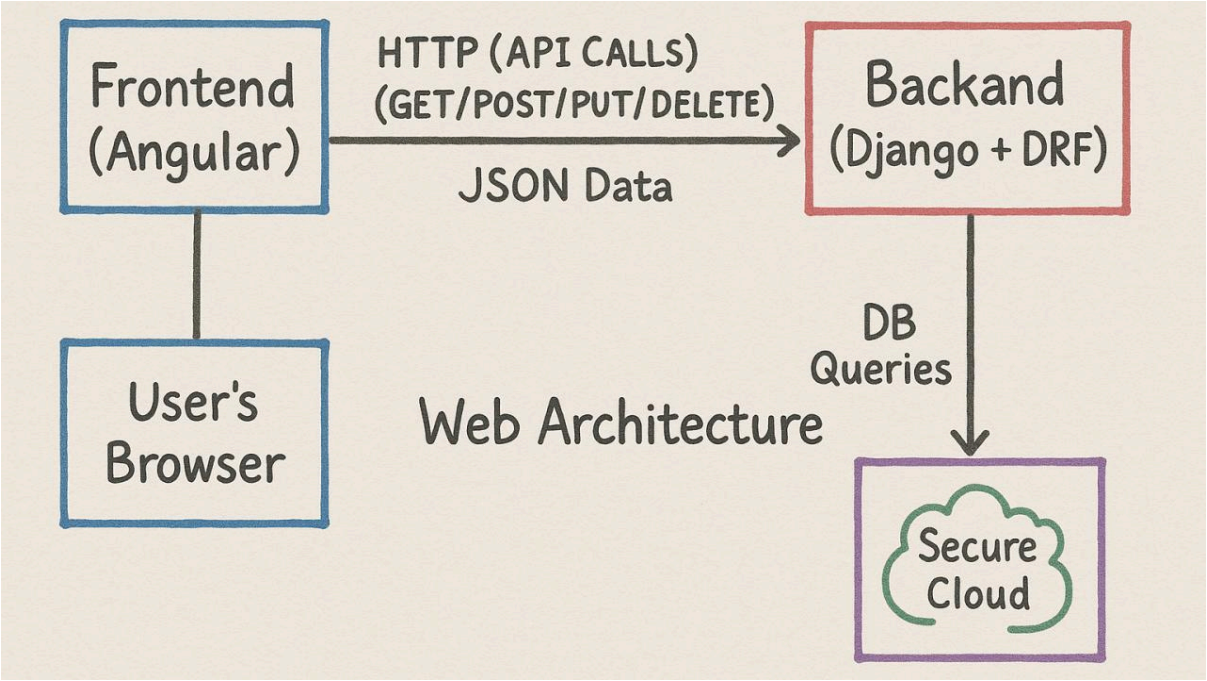
- User registration details (username, email, password)
- Issue reports (title, description, category)
- Status updates (Pending, In Progress, Resolved)

Data is collected through **form submissions** in the frontend and processed via API calls to the backend. The backend validates and stores the data securely in the database.

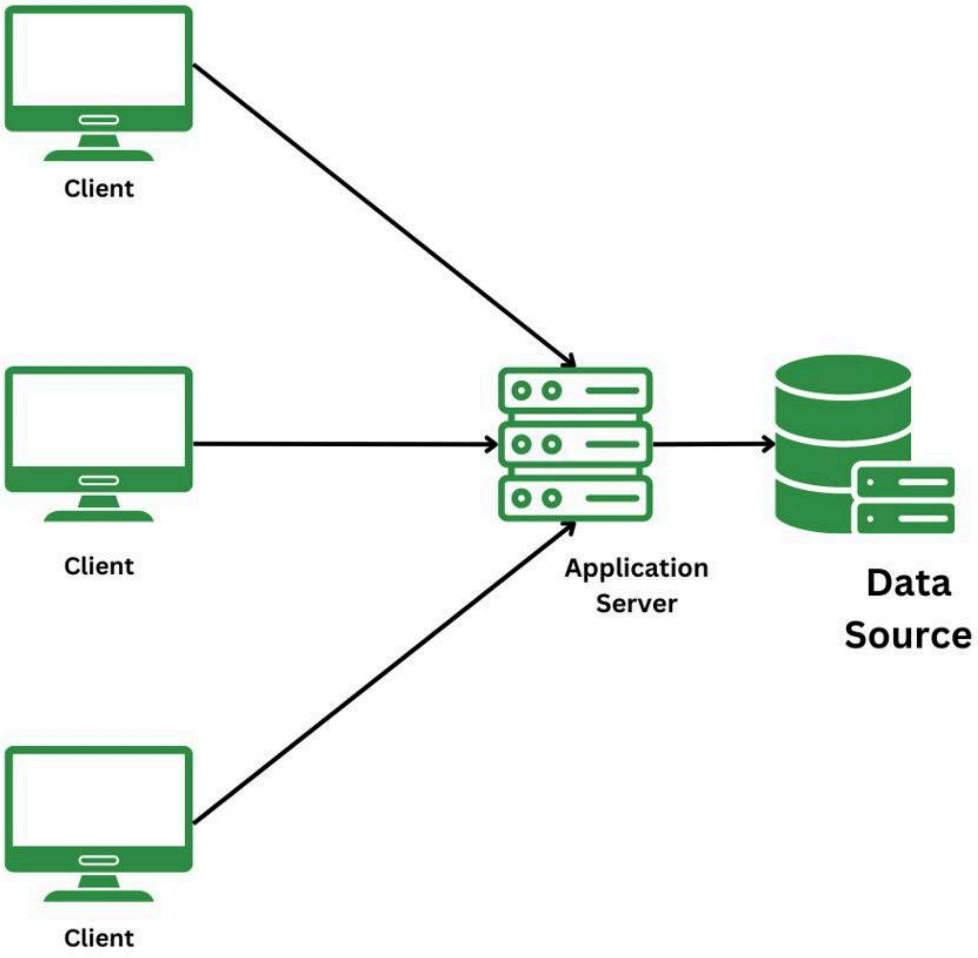
In future enhancements, data collection can be extended to include:

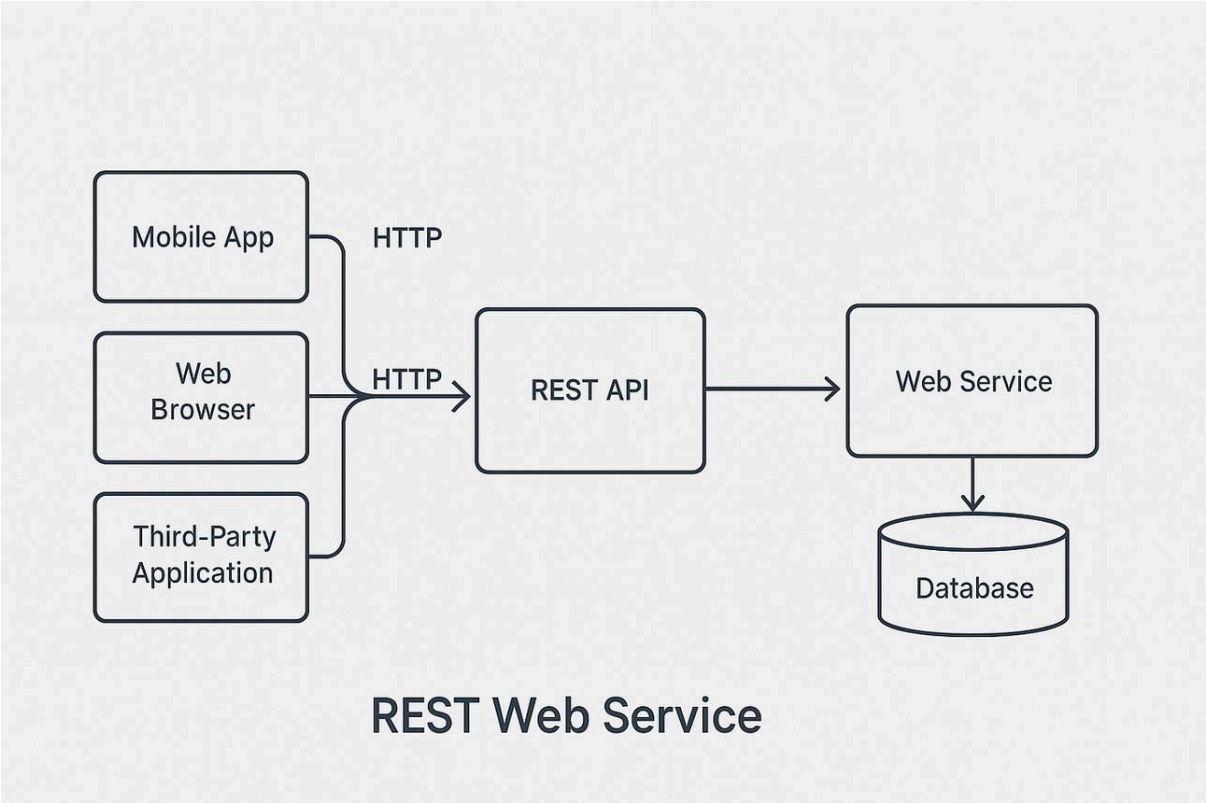
- Image uploads
- Location-based data (GPS)
- Real-time analytics

🏗️ Project Architecture



Three Tier Architecture





🔹 Architecture Description:

The system follows a **three-tier architecture**:

1. Presentation Layer (Frontend)

- Built using Angular
- Provides user interface for interaction
- Sends HTTP requests to backend APIs

2. Application Layer (Backend)

- Developed using Django
- Processes user requests
- Handles authentication and business logic
- Communicates with the database

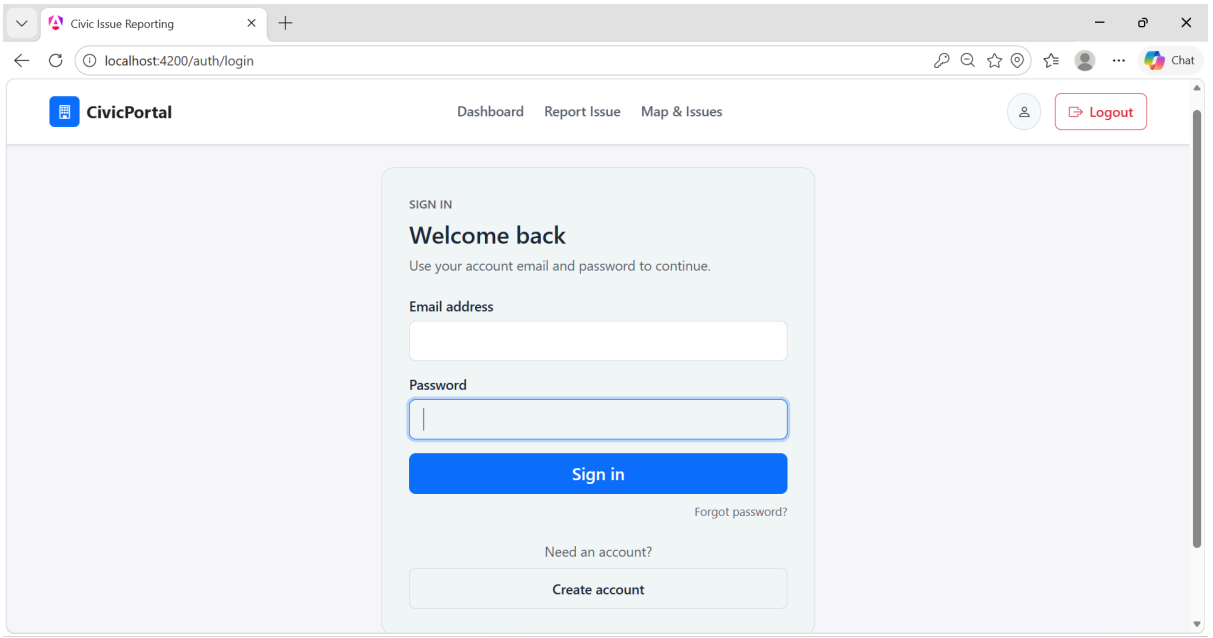
3. Data Layer (Database)

- Managed using PostgreSQL
- Stores all application data
- Ensures data integrity and security

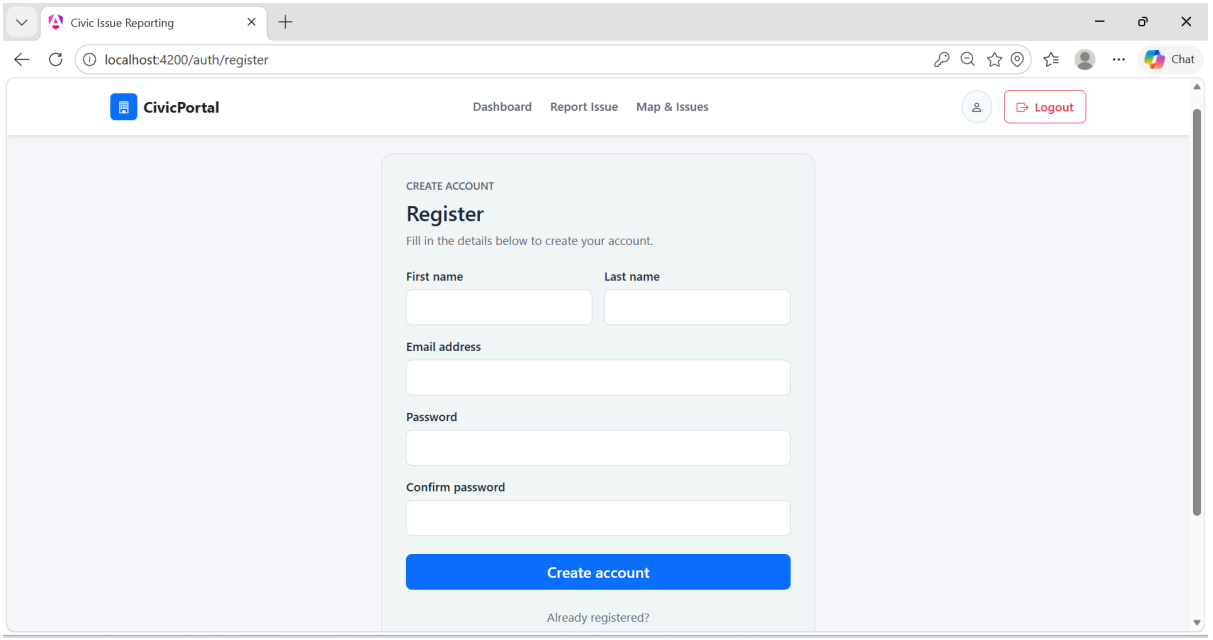
🔄 Workflow:

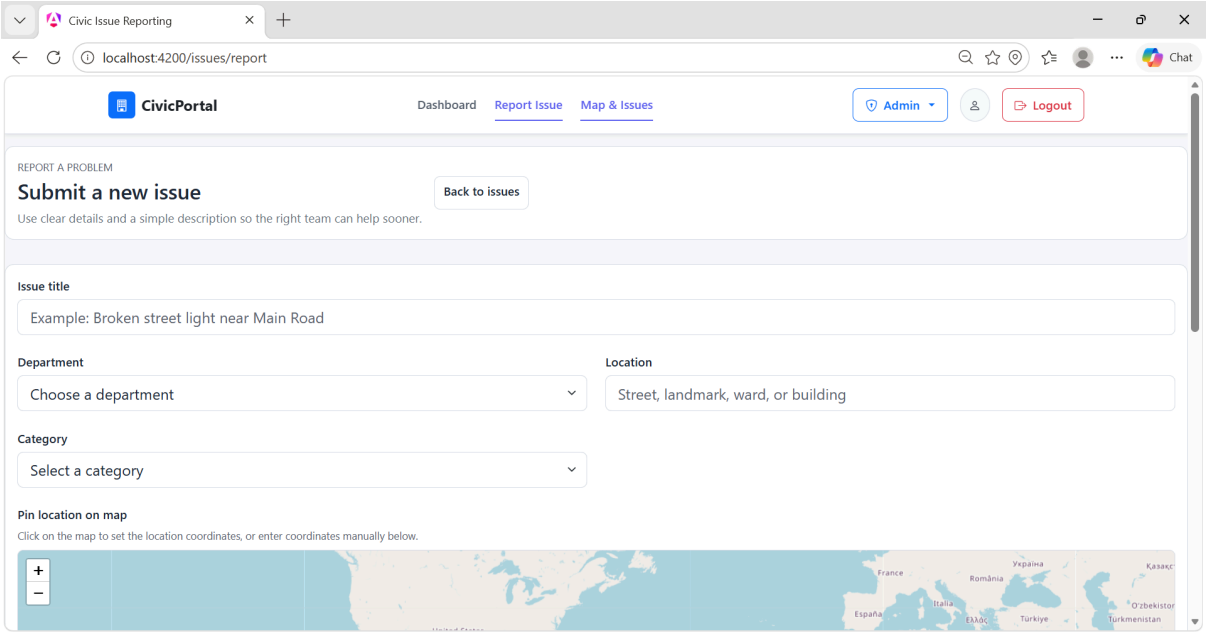
1. User submits a request (e.g., issue report)
2. Angular sends API request to Django
3. Django processes and stores data in PostgreSQL
4. Response is sent back to frontend
5. User sees updated status

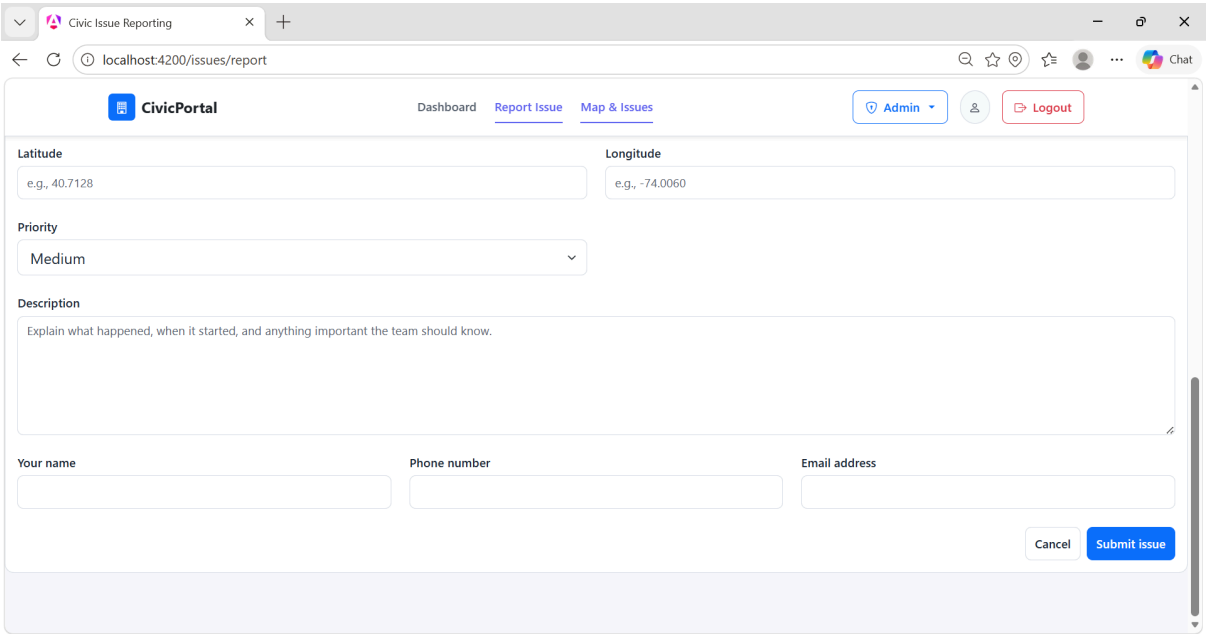
📸Final Project Working (Screenshots + Explanation)



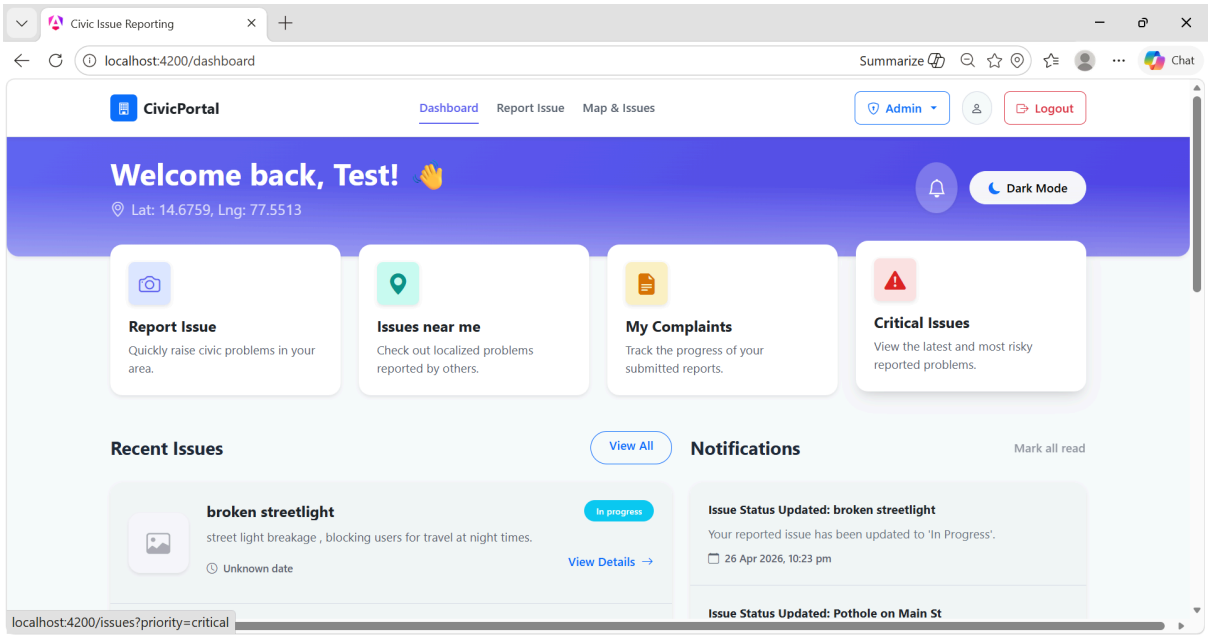
->log in page for authentication and to use essential features of the application

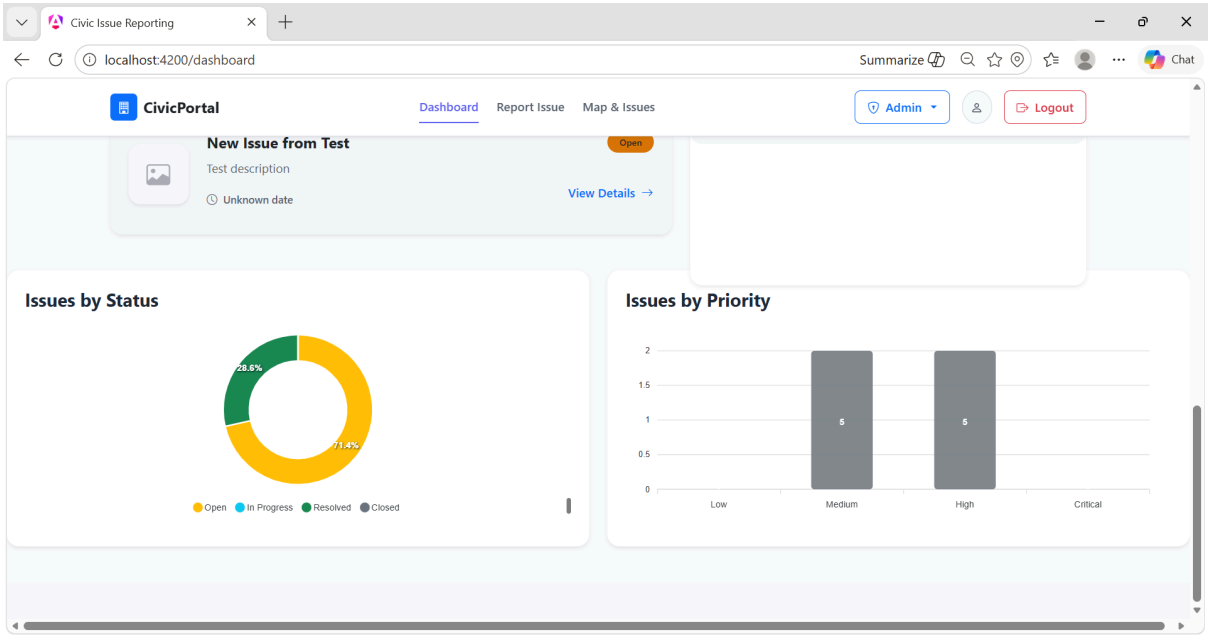




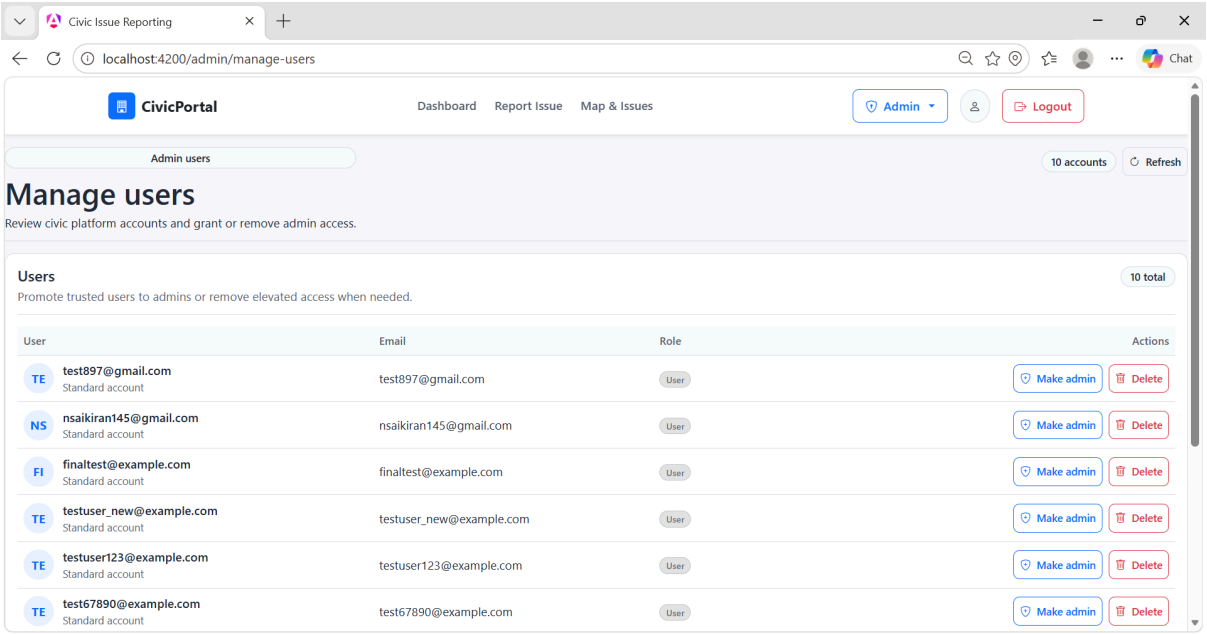


->Issue reporting is easier along with geolocation with latitudes,longitudes.And issues are classified based on the severity.

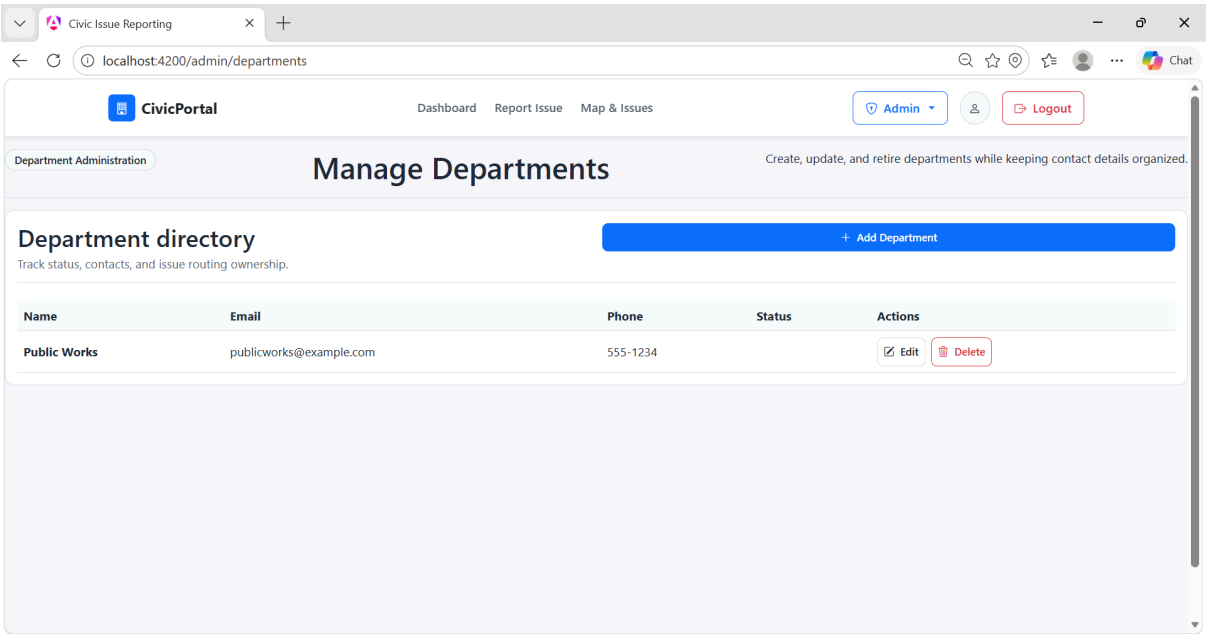




->A simple and clean dashboard for easier understanding of the issues.And to take preventive measures as soon as possible.



->Super admin has authority to make users as admins or remove them permanently.And also has access for setting the status of the issues.



->Admins also have access for managing departments and introducing new departments for future.

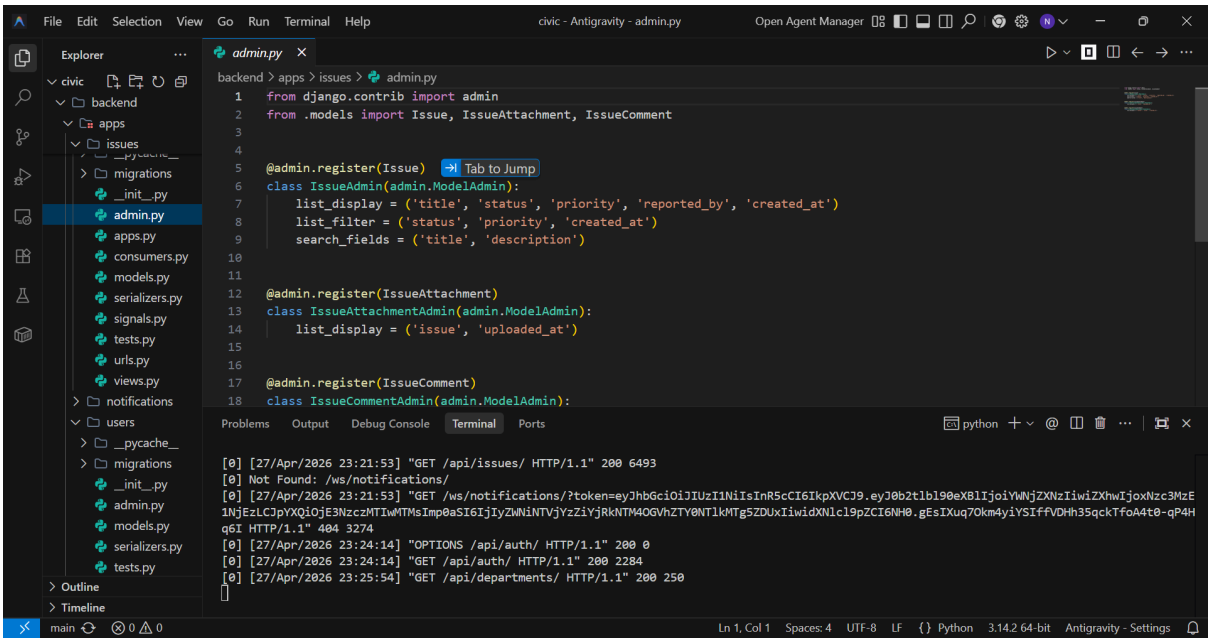
🔹 Key Screens:

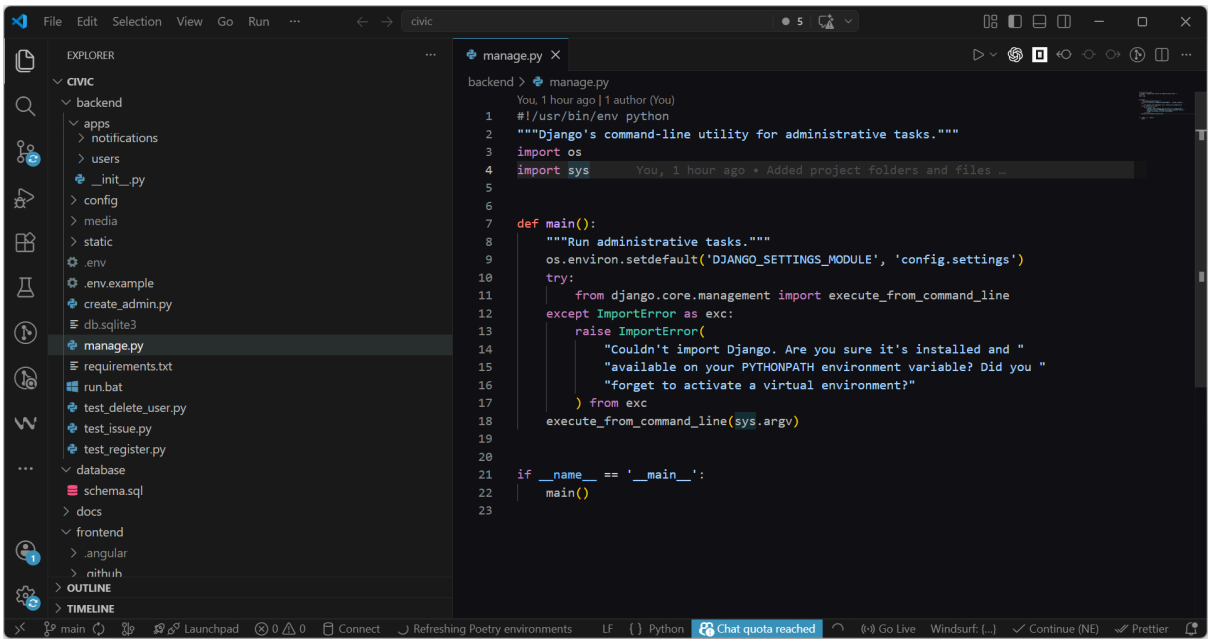
- **Login/Register Page:**
  Allows users to securely access the system
- **Issue Submission Form:**
  Users can report issues with details

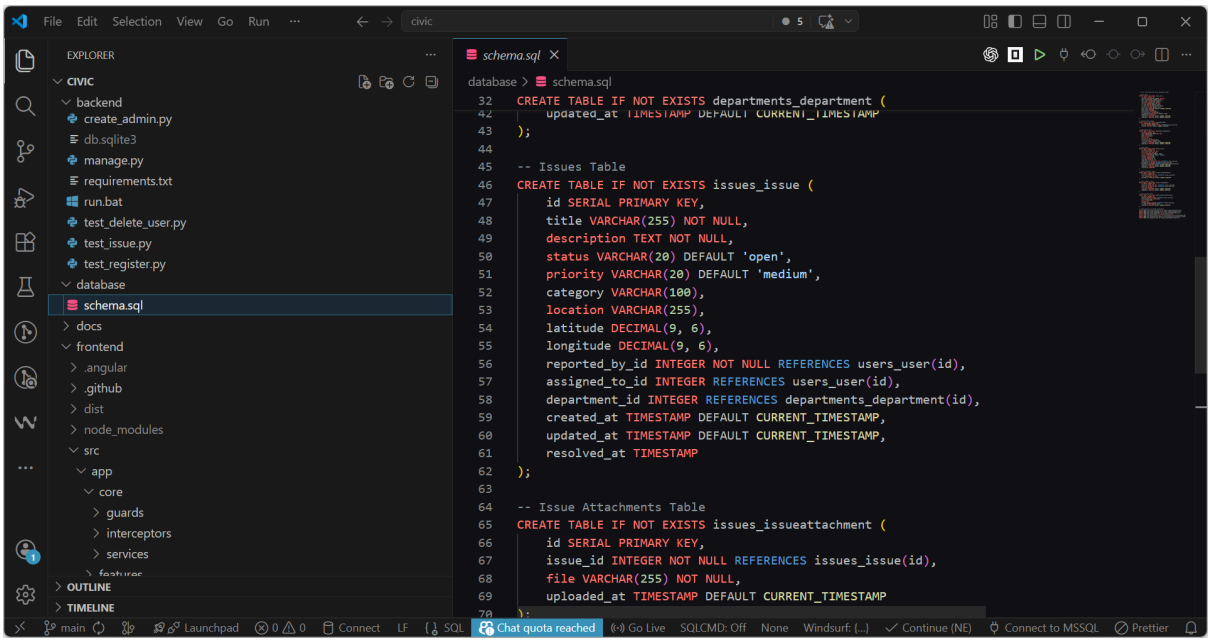
- **Dashboard:**
  Displays submitted issues and their status
- **Admin Panel:**
  Enables authorities to manage and update complaints

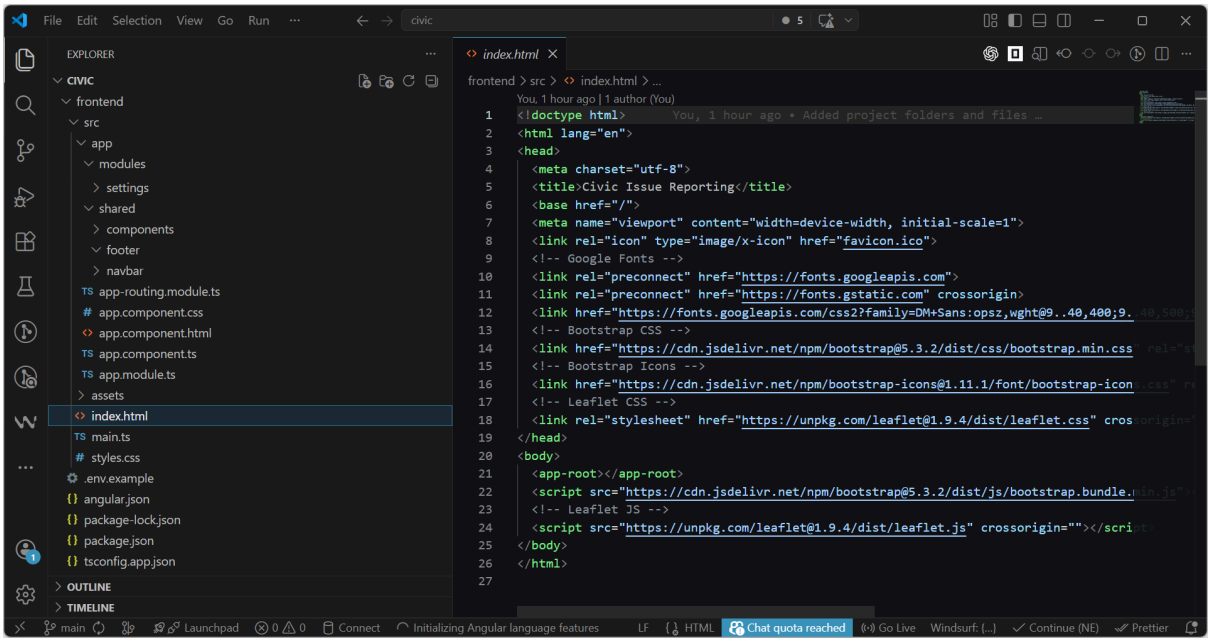
Each module is designed for simplicity and efficiency, ensuring a smooth user experience.

Code screenshots

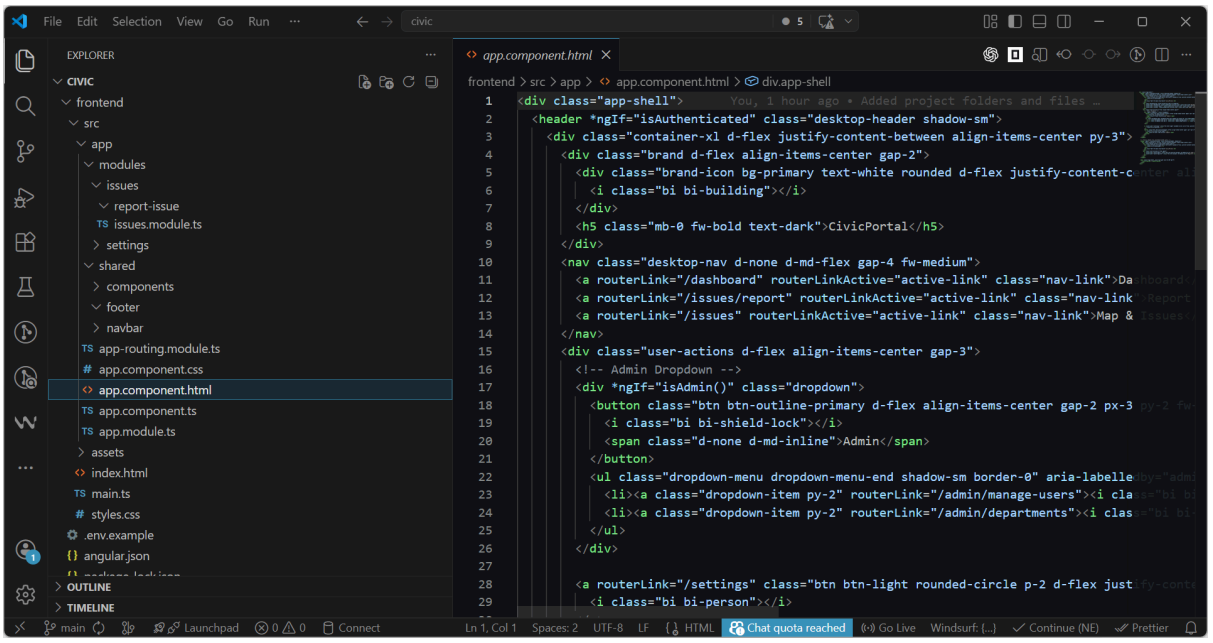








->Main code logic and api routing along with hyperlinks and database setup.



🔗 **Project GitHub Link**

https://github.com/N-Sai-kiran-nsk/Clean-and-Green-Technology-

📈 Results

The project was successfully implemented and tested, achieving the following results:

- Functional web application with frontend and backend integration
- Real-time issue tracking and status updates
- Efficient data storage and retrieval
- Smooth deployment and accessibility via web

The system demonstrates the practical application of full-stack technologies in solving real-world problems effectively.

The project empowers citizens to actively participate in improving their surroundings. Instead of relying on manual complaint methods, users can directly report issues through the web platform. This creates a sense of responsibility and encourages community involvement.

- Useful for Municipalities and Governments

Municipal corporations, local governments, and public departments can use this system to streamline complaint management. It provides a centralized database, real-time status tracking, and administrative dashboards that improve transparency and accountability.

- Can Be Used in Disaster Management Scenarios

The system can be adapted for disaster situations such as floods, fires, earthquakes, or emergencies. Citizens can quickly report incidents, and authorities can coordinate responses more effectively. This makes the platform valuable beyond regular civic management.

- Supports Smart City Initiatives

Modern smart cities depend on digital solutions to improve public services. This project aligns with smart city goals by promoting technology-driven governance, efficient resource management, and improved citizen communication.

Proposed System Features

The proposed **Civic Issue Reporting System** is designed to overcome the limitations of traditional complaint management systems by implementing a modern, scalable, and user-friendly architecture. The system provides faster communication, better transparency, and efficient issue resolution through the following features:

- **Automated Routing:**
  Reported issues are automatically categorized and forwarded to the relevant department based on the type of complaint submitted by the citizen. This reduces manual effort and improves response efficiency.
- **Synchronous and Real-Time Updates:**
  The system provides instant updates to users regarding complaint status changes through WebSocket-based notifications, ensuring continuous communication between citizens and authorities.

- **Modern Single Page Application (SPA) Experience:**
  The frontend is developed using Angular, delivering a smooth and responsive application-like experience without unnecessary page reloads.
- **Data-Driven Insights:**
  Administrators can access dashboards with real-time analytics, charts, and performance metrics that help in monitoring complaints and decision-making.
- **Scalable Backend Architecture:**
  The backend is developed using Django REST APIs, making the system flexible for future expansion such as mobile applications, external integrations, or advanced analytics tools.
- **Secure Authentication and Role Management:**
  Multiple user roles such as Citizen, Staff, Admin, and Superadmin are supported with controlled access permissions.
- **Cloud Deployment Ready:**
  The application can be deployed on platforms such as Vercel and Render for better accessibility and scalability.

🔹 Comaprison Table

Feature/Aspect	Existing Systems	Proposed Civic Issue System
User Interface	Outdated, non-responsive, and difficult to navigate.	Modern, responsive Single Page Application built with Angular.
Tracking Mechanism	Users need to visit offices or make calls for updates.	Real-time status tracking available through user dashboard.
Notifications	Delayed notifications or no updates.	Instant in-app notifications using WebSockets.
Architecture	Traditional tightly coupled monolithic systems	Decoupled frontend and backend using Angular + Django REST API.
Role Management	Limited Admin/User roles only.	Advanced roles such as User, Staff, Admin, and Superadmin.
Data Visualization	Static tables and manual reports.	Dynamic charts, metrics, and dashboard analytics.
Scalability	Difficult to expand or integrate new features.	Easily scalable for mobile apps, AI tools, and services.

Learning and Reflection

The development of this project provided valuable technical and professional learning experiences. It helped in understanding how modern technologies can be combined to create practical solutions.

🔹 Learned Full-Stack Development

The project provided hands-on experience in both frontend and backend development using Angular and Django. This improved understanding of complete application development from user interface to server-side processing.

🔹 Understood API Integration

One of the key learnings was connecting frontend and backend through REST APIs. It helped in understanding request-response cycles, data exchange, authentication, and handling errors.

🔹 Gained Experience in Deployment

Deploying the project on platforms such as Vercel and Render provided practical knowledge about hosting, environment variables, and cloud deployment processes.

🔹 Improved Debugging and Problem-Solving Skills

During development, multiple issues such as configuration errors, API bugs, CORS problems, and database connectivity challenges were encountered. Solving these problems improved debugging techniques, patience, and analytical thinking.

🔹 Personal Reflection

This project demonstrated how technology can be used to solve meaningful real-world problems. It also increased confidence in building scalable applications and working with modern development tools.

🔹 Recap of Objectives and Achievements

The main objective of this project was to design and develop a **web-based platform** for efficient reporting, tracking, and management of civic or disaster-related issues. The system aimed to improve communication between users and authorities while ensuring transparency, faster response time, and better service management.

The project successfully achieved the following objectives:

- Developed a responsive and user-friendly frontend using Angular
- Implemented a secure and scalable backend using Django
- Integrated PostgreSQL for efficient data storage and management
- Enabled users to register, log in, and submit issues through the platform
- Provided real-time complaint status tracking (Pending, In Progress, Resolved)
- Created an administrative dashboard for managing and resolving complaints
- Successfully deployed the application using Vercel and Render

The completed system demonstrates how modern web technologies can be used to solve practical real-world problems effectively.

🔹 Future Scope of the Project

Although the project meets its current goals, several enhancements can further improve its functionality and scalability.

📱 Mobile Application Development

A dedicated mobile application for Android and iOS can be developed to make reporting faster and more convenient.

🤖 AI-Based Issue Prioritization

Artificial Intelligence can be integrated to automatically classify complaints and prioritize urgent cases based on severity.

🗺️ GPS and Map Integration

Location services can be added to allow users to mark exact issue locations on a map, improving field response efficiency.

🔔 Notification System

SMS, email, or push notifications can keep users informed whenever complaint statuses are updated.

📊 Advanced Analytics Dashboard

Analytical reports and charts can help authorities study trends, response times, and resource usage for better planning.

☁️ Large-Scale Deployment

The system can be upgraded for city-wide or state-wide implementation with stronger security, higher scalability, and multi-department integration.

🔹 Final Conclusion

This project provides a reliable and scalable solution for digital issue management. It improves efficiency, accountability, and citizen engagement while creating a strong foundation for future smart governance systems. With additional enhancements, the platform can evolve into a comprehensive solution for municipalities, governments, and disaster response agencies.